

HIMDRISHTI

Polar Voyage Decision-Support & Risk
Assessment

SIH 2026 PROTOTYPE - NOT FOR
OPERATIONAL USE

Voyage Plan & POLARIS Dossier
HMDR/VYG/01

VOYAGE PLAN & POLARIS DOSSIER

System-generated assessment based on the submitted voyage configuration and available HimDrishti scientific pipeline outputs.

PROGRAMME Smart India Hackathon 2026	PROBLEM STATEMENT SIH 2026 · PS 26059	DOCUMENT ID HMDR-20260929-1033-001
GENERATED 29 Sep 2026, 10:33 IST	ASSESSMENT REGION Southern Ocean (60°S–90°S)	RECORD STATUS Prototype assessment record

01 Executive Summary

VOYAGE ASSESSMENT

HimDrishti evaluated the configured Cape Town to Bharati voyage using the available deterministic routing and POLARIS risk-assessment pipeline outputs. The generated assessment is intended to support human review and does not constitute an operational navigation instruction.

02 Voyage Particulars

Origin	Cape Town, South Africa	Destination	Bharati Station, Larsemann Hills
Departure	29 Sep 2026, 10:33 IST	Vessel	NCPOR Charter Vessel
Draft	7.5 m	Speed	12 kn
Operating Mode	BALANCED	Risk Budget	Post-search evaluation

03 Route Assessment

Recommended Route: Cape Town → Southern Ocean → Bharati Station

DISTANCE 4,860 nm	TRANSIT TIME 405.0 h	RISK COST 18.4	DISTINCT PATH No
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Route convergence: The recommended route converges with an alternative search under the current synthetic hazard field; the system does not apply an artificial geometric offset.

04 POLARIS Risk & Compliance

RISK SCORE 18 / 100 Computed from available POLARIS risk outputs.	RISK LEVEL LOW RISK Based on currently available route-level risk outputs.	RIO CATEGORY Category II POLARIS-derived category where available.
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05 Route Geometry Trace

Segment	Method / Source	Record
Approach	Great-circle approach to pipeline grid entry	Cape Town to model-domain entry
Pipeline	Deterministic A* route search	POLARIS-aware path through model grid
Arrival	Destination approach / snap	Model-domain exit to Bharati destination

Route geometry shown here is the geometry returned by the current voyage session. No synthetic offset is applied merely to make routes appear different.

06 Alerts & Data Provenance

Type	Message / Source	Time
ICE	Route enters elevated sea-ice assessment zone; review current operational ice information before departure.	09:58 IST
ROUTE	Recommended and alternative route paths converge in a model segment.	10:02 IST
SYSTEM	Scientific pipeline outputs successfully attached to the voyage session.	10:03 IST

Dataset / Component	Record / Status	Retrieved / Computed
POLARIS Risk Model	Route-level risk evaluation	29 Sep 2026, 10:02 IST
Sea-Ice Field	Model grid used by routing pipeline	29 Sep 2026, 10:02 IST
Iceberg Ensemble	Available drift-risk component	29 Sep 2026, 10:02 IST
Voyage Session	Current session input and route state	29 Sep 2026, 10:03 IST

07 Captain / Master Decision

RECORDED DECISION	Final voyage and navigation authority
ACCEPTED	Captain / Master remains responsible for the final decision.
Remarks: Route accepted for prototype demonstration and human review.	

08 Audit Trail

Time	Event	Detail
10:02:14 IST	ROUTE_GENERATED	Three route strategies evaluated by the backend pipeline.
10:03:01 IST	CAPTAIN_DECISION	Recommended route accepted; decision recorded in session audit trail.

Prototype and use limitation: This document is a demonstration output of the HimDrishti SIH 2026 prototype. It is not an official Government of India publication, not an operational navigation instruction, and not a substitute for authoritative charts, forecasts, ice information, vessel procedures, or the professional judgment of the Captain / Master.